

Visual Hypothesis Testing: The Effect of Visualisation on the Statistical Power of the Lineup Protocol

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Abstract

This report investigates the impact of visualisation choice on the statistical power of the lineup protocol introduced by Buja *et al.* (2009)[1], a framework for placing a human observer as a test statistic. Using the hypothesis test for equal variance, we constructed lineups of boxplots, density plots, violin plots and absolute density plots at two effect sizes ($\sigma = 1.2$ and $\sigma = 1.7$) and administered a survey to 47 students and faculty from the School of Mathematics, to investigate their detection accuracy.

Detection accuracy varied substantially across plot types, with density and absolute density plots achieving mean accuracies of approximately 81% and 82% respectively, violin plots achieving 49%, and boxplots performing most poorly at 23%. Logistic regression confirmed that plot type is a highly significant predictor of correct identification, while academic experience had no statistically significant effect. Effect size was also found to be a significant predictor of accuracy, though no significant interaction between plot type and effect size was found.

Qualitative analysis of open-ended responses revealed that higher-accuracy participants tended to report cues related to spread and distributional shape, whereas lower-accuracy participants more often cited separation and overlap. These findings suggest that density-based encodings, which translate variance into positional judgements, are the most effective visualisations for variance-based visual inference, consistent with Cleveland and McGill's hierarchy of graphical perception tasks[2].

Declaration

We declare that this thesis was composed by ourselves and that the work contained therein is our own, except where explicitly stated otherwise in the text. We declare that the following Generative AI tools were used in the completion of this work: ChatGPT, for debugging code in sections 4 and 5 of the report; ChatGPT for spelling and grammar throughout the report.

(Matthew Duncan, Frank Fan, Loizos Theodorou)

Dedicated to our supervisor, Simon Taylor, in grateful recognition of his support, patience, and guidance throughout this project. We also dedicate this work to our project group, whose collaboration, commitment, and shared effort made this research possible, and to all of our friends and colleagues who participated in our survey.

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Chapter 1

Introduction

Data visualisation is a widely used tool for identifying patterns in data. Visualisation methods such as scatter plots, boxplots and histograms allow the viewer to detect relationships, trends and discrepancies that may be difficult to identify through numerical summaries alone. However, while these tools are very powerful for exploring data, their interpretive nature makes them inherently subjective. Two different people may draw two separate conclusions from the same image. This raises an important question: can data visualisations be used not only for exploring the data, but also to draw conclusions about the structure of the data?

Visual hypothesis testing aims to bridge the gap between exploratory data analysis and statistical inference by formalising the process of drawing statistical conclusions from visual evidence. This study investigates the effect of different visualisation techniques on the statistical power of these hypothesis tests.

Humans are able to visually identify subtle patterns that conventional numerical tests may overlook, and a systematic protocol can establish statistical validity for these observations. For instance, we will discuss how Buja et al. (2009) [1] considered a plot as a test statistic and used human pattern recognition to determine significance. The introduction will also delineate the objectives of our project: to assess the impact that factors such as the choice of visualisation and background and experience of the subject can have on the power and validity of a visual hypothesis test. We will provide a concise overview of the methodology (permutation-generated null plots and human evaluation) and outline the principal questions:

- Are observers able to accurately identify the data plot? We expect observers to perform above chance across all plot types, with detection accuracy highest for density-based encodings.
- How does this visual representation compare to traditional assessments? We expect visual inference to achieve comparable power to numerical tests at larger effect sizes, with performance degrading more sharply than numerical methods as the effect size decreases.
- What impact does statistical experience have on visual inference? We expect that statisticians with more education will have greater insight into visual cues.

The lineup protocol provides such a framework by treating the plot as a test statistic and replacing a numeric reference distribution with a page of null plots. Mimicking a police lineup [1], we generate $n - 1$ null plots under null hypothesis H_0 , then randomly insert the actual data plot at a random position under a veil of ignorance (neither the observer nor the analyst knows the location of the real graph) and request independent viewers to identify the plot that appears most different.

The remainder of this report is organised as follows:

- **Section 2: Background** provides the theoretical foundation, discussing the subjectivity of exploratory data analysis and the formal protocols introduced by Buja et al. (2009) [1].
- **Section 3: Methods** outlines the experimental design, including the construction of lineups and the recruitment of participants from the School of Mathematics.
- **Section 4: Survey Results** examines the data from our study, analysing detection accuracy and the impact of plot type on statistical power.
- **Section 5: Qualitative and Quantitative Analysis** explores the relationships in the survey data between qualitative aspects such as visual cues reported in the open-ended response, and quantitative analysis such as model fitting to explore the strength of predictors.
- **Section 6: Conclusion** considers possible implications and limitations of our research and considers various avenues for future investigation.

Chapter 2

Background

Exploratory data analysis (EDA) and model diagnostics (MD), as introduced by Tukey (1977) [9], emphasise the role of graphical methods in informing about the structure of the data before (EDA) and after (MD) the model is fitted. EDA and MD are seen as data analyses that are not completely reliable due to the subjectivity of human judgment, resulting in a wrong interpretation of the data. Therefore, without a formal inferential framework, the patterns seen through visual inspection could not be assessed with much statistical certainty. The main challenge, then, is to maintain the advantages of human visual judgment while also introducing these interpretations in solid statistical reasoning.

This subjectivity can be clearly seen in the two plots shown in Figure 2.1. The plot on the left displays a scatterplot of Y against X , which is part of the EDA since the model has not been fitted yet. At first glance, one might conclude that there is no relationship between the variables because the points appear to be widely scattered with no obvious pattern. However, another viewer may notice a slight upward tendency in the points as X increases and interpret this as evidence of a weak positive trend. Thus, the same plot can lead to different interpretations depending on who is looking at it.

The plot on the right in Figure 2.1 is a Residuals *vs* Fitted plot, which is used in MD to evaluate how well a fitted regression model captures the underlying relationship in the data. If the model is appropriate, the residuals should be randomly scattered around the zero line. In this case, one viewer might consider the residuals to be sufficiently random and therefore judge the model to be adequate. However, another viewer may notice a slight curved pattern in the residuals and interpret this as evidence that the model is missing a nonlinear component. The first viewer would keep the model as is, while the second would modify or refit the model.

One way that is used to address the subjectivity of human interpretations in EDA and MD is to add measures of uncertainty in visualisations. This can be established using confidence bands. Confidence bands show the amount of variability and uncertainty in an estimated relationship by displaying a region around the fitted curve where the true relationship is likely to lie with a certain probability. Adding confidence bands to a curve can produce some advantages. They show uncertainty directly on the plot, which can help reduce the risk of misinterpretation of some patterns that may simply be due to sample variation. In

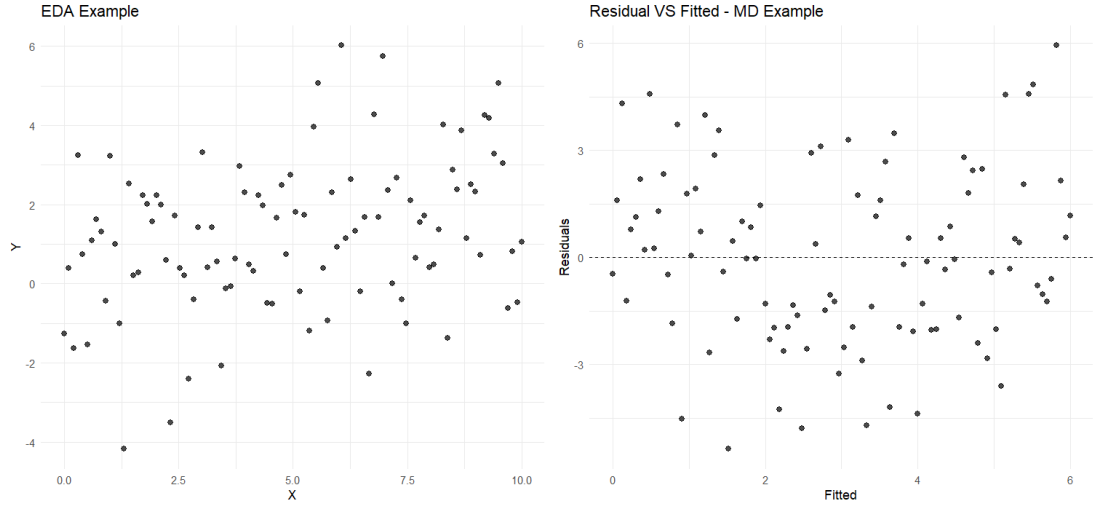


Figure 2.1: EDA and MD plot examples of subjectivity.

addition, confidence bands add some statistical contexts to the graphical analysis.

Figure 2.2 shows this idea by adding confidence bands in the EDA and MD examples used in Figure 2.1. The confidence bands in the EDA example plot show that the upward relationship of the fitted line between X and Y is quite weak and uncertain. This makes it difficult to draw conclusions only using graphical inference. In the MD example plot, the confidence bands make it easier to notice that the residuals have a slight curved pattern. This suggests that the fitted model may not fully capture the relationship in the data and could indicate that a nonlinear component is missing from the model. However, the confidence bands also suggest that the curvature found is small compared to the overall uncertainty, which means that the evidence for a missing relationship is not very strong. As a result, different viewers may still disagree on whether the model should be modified.

Thus, from the previous example we can conclude that, despite the benefits, confidence bands do not remove subjectivity and can introduce new types of distraction. Adding shaded areas and smooth curves can draw attention away from the original data points, making it more difficult to notice patterns, outliers, or differences in the spread of the data. Furthermore, they make interpretation more demanding, since the viewer now has to judge how the observed patterns relate to the amount of uncertainty shown. Therefore, confidence bands tend to shift the subjectivity of visual interpretation rather than minimising it, suggesting the challenge to develop methods that assess visual patterns more formally.

2.1 Formalising Visual Inference

Buja *et al.* (2009) [1] addressed this challenge by proposing a formal inferential framework for data visualisation, positioning the human observer as a test statistic. Their core contribution was to formalise visual judgments in terms analogous to hypothesis testing. If a viewer can reliably distinguish the observed data plot from plots generated under a null hypothesis, this is taken as evidence against the

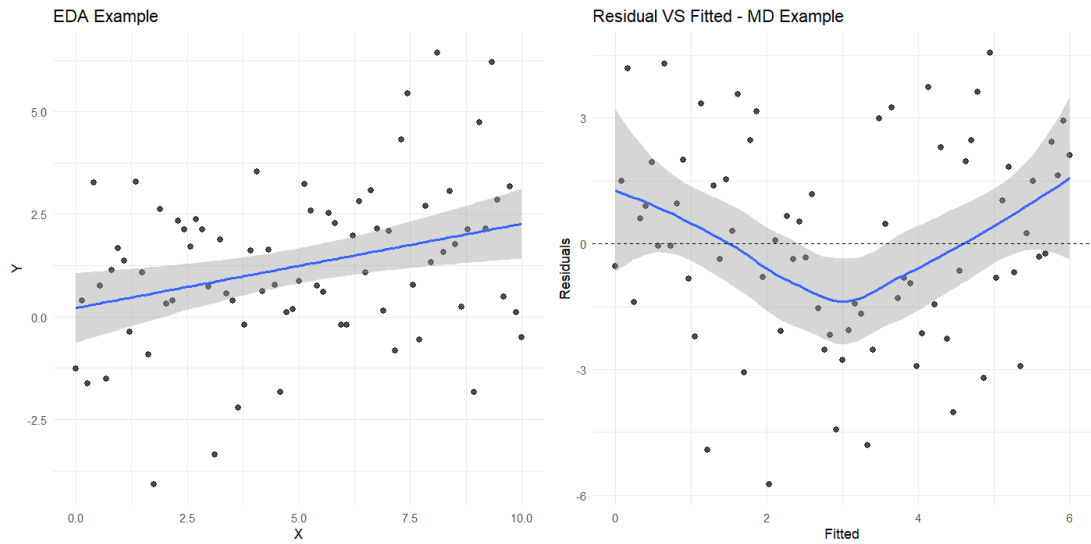


Figure 2.2: The same EDA and MD plot examples of subjectivity as Figure 2.1 but with confidence bands.

null, just as with a numerical test statistic. To operationalise this idea, Buja *et al.* introduced two complementary protocols: the lineup protocol and the Rorschach protocol.

More generally, Buja *et al.* (2009) [1] aimed to connect visual inference with standard statistical inference. They matched many familiar ideas from the standard method of hypothesis testing with the corresponding components in visual analysis. For example, a numerical test statistic is replaced by a plot and the null distribution is replaced by a variety of plots generated under the null model. In this case, instead of rejecting the null hypothesis using a numerical value, the viewer is rejecting the null by identifying the plot that looks the least like the others. These ideas are commonly presented in a table like Table 2.1 that matches standard statistical concepts with their visual equivalents. By making these parallels explicit, Buja *et al.* show that visual hypothesis testing follows the same logic as the classic method, rather than being a secondary tool or a subjective process.

2.2 Rorschach Protocol

Before introducing a formal visual inferential framework, such as the lineup protocol, it is necessary to examine the reliability of the human observer. As discussed earlier, the graphical interpretation of EDA and MD is subjective. Rather than attempting to suppress this subjectivity, Buja *et al.* (2009) [1] proposed a way to measure it directly.

Therefore, they introduced the Rorschach protocol. The name is given metaphorically after the work of Hermann Rorschach, the Swiss psychiatrist who created the famous inkblot test. It is named like this, since the protocol is designed to measure the viewer's tendency to perceive structure in randomness.

In this protocol, the analyst is shown a series of plots that are generated from

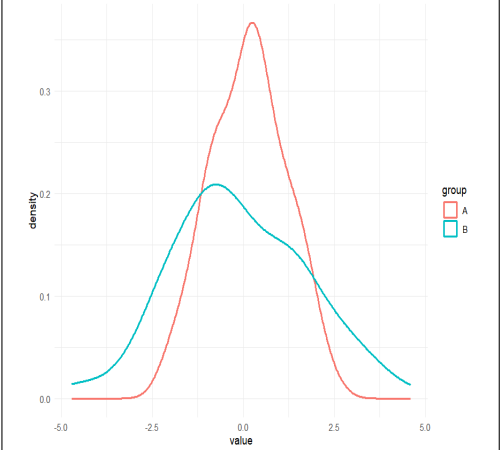
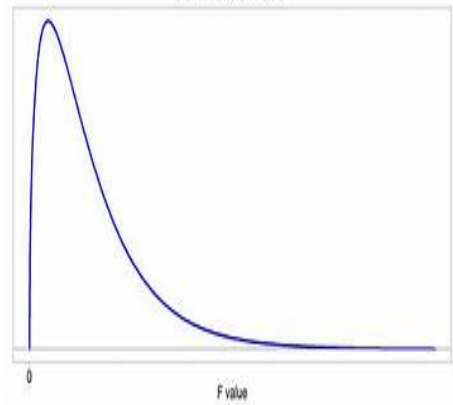
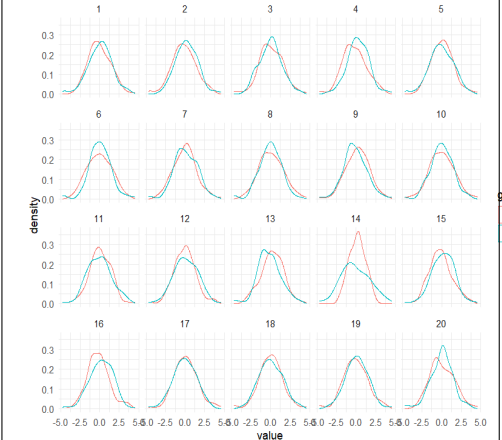
	Conventional inference	Lineup protocol
Hypothesis	$H_0 : \sigma_X^2 = \sigma_Y^2$ vs $H_1 : \sigma_X^2 \neq \sigma_Y^2$	$H_0 : \sigma_X^2 = \sigma_Y^2$ vs $H_1 : \sigma_X^2 \neq \sigma_Y^2$
Test statistic	$T = \frac{S_X^2}{S_Y^2}$	
Sampling distribution		
Reject H_0 if	Actual T is extreme	Plot of true data is identifiable

Table 2.1: Comparison between conventional inference and visual inference [7].

the null hypothesis. Therefore, there is no true signal from the construction of these plots. Hence, any declaration of structure in these null plots is necessarily spurious. Then, the protocol records the frequency of these discoveries and estimates empirically the analyst’s Type I error rate (the probability of falsely detecting at least one pattern when none exists).

For the protocol to be performed properly, the roles need to be clearly separated. A protocol administrator is responsible for generating the null data and creating the corresponding plots, while the analyst only evaluates the plots. To reduce bias, neither the administrator nor the analyst should know whether a plot based on the true data has been included among the null plots.

In practice, implementing the Rorschach protocol would mean adding an initial step to the data analysis process: before viewing the real data plot, the analyst first examines a series of null plots. This would allow analysts to calibrate their judgment of random variation to reduce the risk of misinterpreting weak patterns. This calibration step prepares the ground for more formal visual testing procedures, such as the lineup protocol discussed next.

2.3 Lineup Protocol

The lineup protocol serves as a bridge between exploratory data analysis and formal statistical inference by treating a plot as a test statistic (as shown in Table 2.1). This method was formally validated by Majumder *et al.* (2013) [7], who demonstrated that the probability of an observer correctly identifying the data plot among null alternatives can be used to calculate a visual p -value, providing a rigorous foundation for the human as a test statistic approach used in this study.

A lineup consists of n plots, one of which displays the real data, and the remaining n are null plots generated under the null hypothesis. In many applications, these null plots are produced by permuting the values of the observed data (for example, by randomly reassigning the response variable relative to the predictor). The real plot is then placed at a random position among the null plots, so the viewer must rely on visual evidence rather than prior knowledge when attempting to identify it. The result is an inferentially valid p -value. For $n = 20$, if the viewer selects the plot of the real data among the 20 different plots, then the discovery can be assigned a p -value of $0.05 = \frac{1}{20}$ and there is evidence to reject the null hypothesis. The lineup protocol, therefore, directly mirrors the logic of classical hypothesis testing (Table 2.1), but replaces numerical measures with human visual perception.

An important feature of this protocol is that the plots are arranged randomly. Moreover, when H_0 is true, the real plot is generated under the same assumptions as the null plots, so all plots are exchangeable. This makes the probability of selecting each plot the same ($\frac{1}{20}$). Therefore, even someone with high experience in data analysis should not be able to identify the real plot more often than by chance if there is no actual structure in the data. In this way, the Type I error rate is controlled by the design of the lineup itself. This is different from the Rorschach protocol, which only measures how frequently an analyst tends to see patterns in random variation, but does not control that tendency.

Furthermore, according to Table 2.1 seen before, instead of calculating a numerical test statistic, viewer perception plays the role of the test statistic (using the lineup protocol). The null distribution is represented by the collection of null plots. Rejecting the null hypothesis corresponds to successfully identifying the real data plot among them.

To illustrate this idea, Figure 2.3 shows an example of the lineup protocol for a correlation test. One of the 20 plots displays the true data, while the other 19 are random null plots generated under the null hypothesis by permuting X and Y randomly. Under the null hypothesis, the points of the plot should be randomly scattered without a pattern. On the other hand, if a plot is observed to be the least like the others, then it should be the plot of the true data. If a viewer correctly identifies this plot, the null hypothesis of no relationship can be rejected at the 5% significance level.

In the case of Figure 2.3, plot number 5 represents the true data. This plot stands out because it shows a clearer positive linear trend between X and Y , while the other plots mostly display random scatter without a systematic direction. Thus, this makes it visually distinguishable and the correct selection.

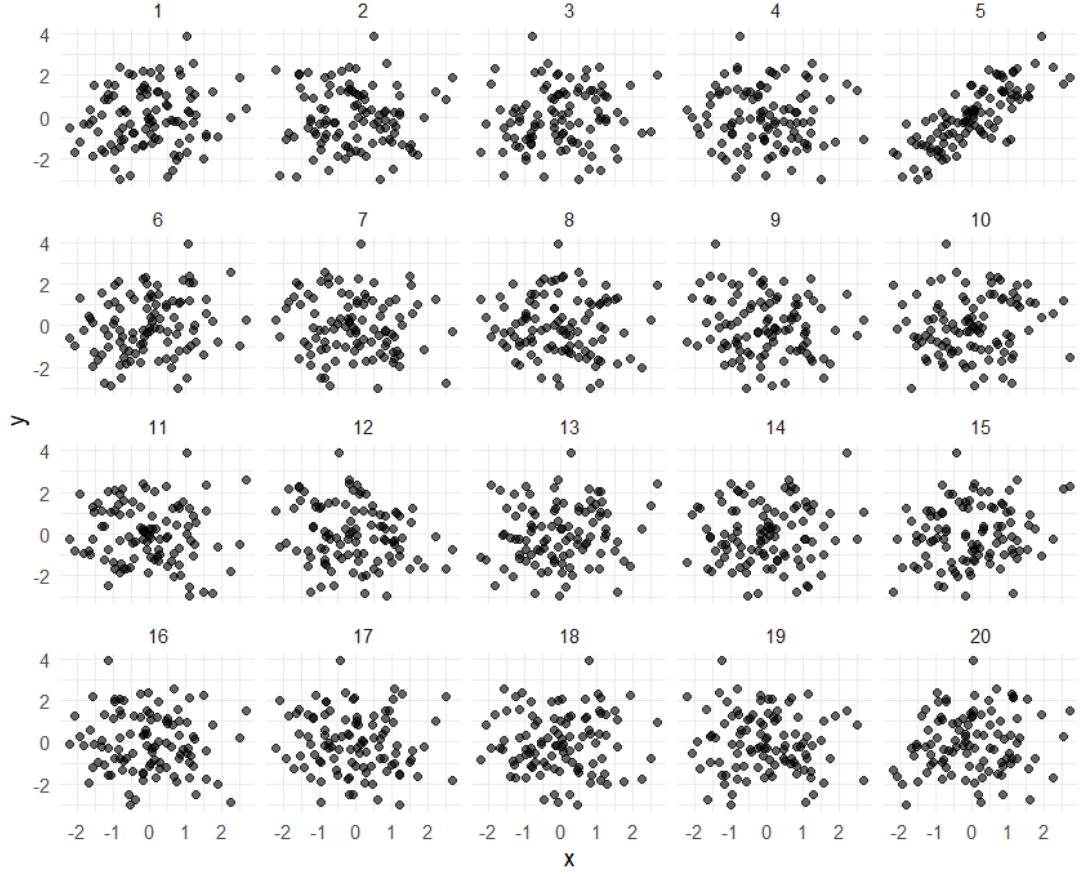


Figure 2.3: Example of the lineup protocol for a correlation test.

This example illustrates how the lineup protocol addresses the issue of subjectivity discussed earlier in this chapter. In EDA and MD, a single plot can lead to different interpretations because there is no clear reference for what random variation looks like. The lineup protocol reduces this problem by providing an explicit comparison. The choice of the viewer is now relative to 19 other plots which are generated under the null hypothesis. Hence, using the lineup protocol can create a formal inferential framework for data visualisation.

2.4 Power

Building on the inferential framework, it needs to be stated that its effectiveness also depends on its statistical power. In this context, power is defined as the probability that the observer correctly identifies the true data plot, leading to rejecting the null hypothesis. Although the inclusion of null plots reduces arbitrary interpretation, accurate identification of the true plot still depends on how clearly the signal stands out from random variation. Larger deviations from the null hypothesis generally produce patterns that are more visually distinct. Thus, they increase the probability that the observer selects the correct plot.

As discussed previously, under the null hypothesis, the probability that an

observer correctly identifies the plot of the true data (in each lineup of our survey) is $p_0 = \frac{1}{20} = 0.05$. Furthermore, assume that K observers evaluate a lineup and let X be the number of those observers that correctly identify the true data. This number follows a binomial distribution

$$X \sim \text{Binomial}(K, p_0).$$

Using the definition proposed by Majumder *et al.* (2013) [7], a rejection rule for the lineup test can be defined by determining the smallest integer x_α (number of the least observers correctly identifying the true panel) such that

$$\mathbb{P}(X \geq x_\alpha | H_0) \leq \alpha,$$

where $\alpha = 0.05$ is the desired significance level. If the number of correct identifications is above that threshold, we reject the null hypothesis.

Hence, to estimate the power of the lineup test, we first calculate the probability that an individual participant correctly identifies the true panel (for a given visualisation and effect size). This is given by [7]

$$\hat{p} = \frac{X}{K}.$$

Therefore, treating $\hat{p}$ as the underlying probability of correct identification, the probability that the lineup test rejects the null hypothesis (the power of the visual test) can be written as [7]

$$\text{Power} = 1 - \text{Binomial}_{K, \hat{p}}(x_\alpha - 1),$$

which represents the probability that at least x_α observers correctly identify the true panel when the probability of a correct identification is $\hat{p}$.

2.5 Plot Types

In this section, we introduce the types of plots that will be used in the following analysis. These visualisations form the basis for the lineup tests that we will describe in the next chapters.

2.5.1 Boxplot

The boxplot, introduced by Tukey (1977) [9], is a visual summary of a univariate distribution based on five summary statistics: the minimum, the first quartile Q_1 , the median Q_2 , the third quartile Q_3 , and the maximum. Given a sample $x_1, x_2, \dots, x_n$ arranged in ascending order, the quartiles are defined as the 25th and 75th percentiles of the empirical distribution, respectively, and the interquartile range is given by

$$\text{IQR} = Q_3 - Q_1. \tag{2.1}$$

The box spans from Q_1 to Q_3 , with a horizontal line drawn at the median Q_2 . The whiskers extend from the edges of the box to the most extreme observed

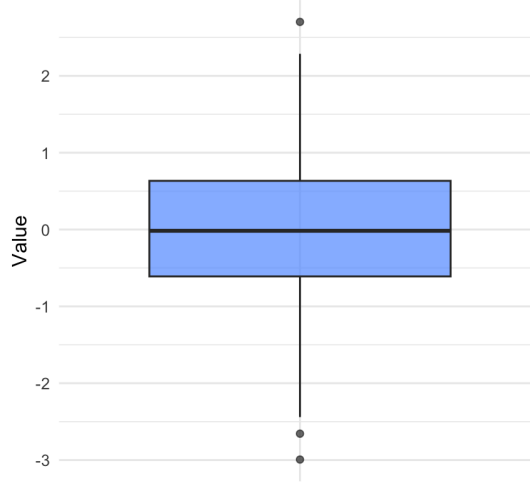


Figure 2.4: Example of a boxplot.

values that fall within the fences

$$[Q_1 - 1.5 \times \text{IQR}, Q_3 + 1.5 \times \text{IQR}]. \quad (2.2)$$

Observations falling outside these fences are plotted individually as outlier points. The boxplot therefore encodes spread through the length of the box and the reach of the whiskers, both of which are judgements of length along a common scale. Cleveland and McGill (1984)[2] rank such position and length comparisons highly in their hierarchy of graphical perception tasks, suggesting that boxplots should in principle support accurate visual inference. However, because the distribution is reduced to five summary statistics, features such as multimodality, skewness, and tail behaviour are not directly encoded, which can limit the information available to a viewer attempting to detect differences in variance.

2.5.2 Density Plot

A density plot displays a smooth estimate of the probability density function of a univariate sample. The oldest and most widely used method for distributional assessment is the histogram, which places observations into fixed-width bins and counts observations that fall in each bin. However, as noted by Silverman (1986)[6], the histogram can be quite substantially improved due to its inefficient use of the data and potential disruption of the true patterns present in the data. These drawbacks motivate the use of the kernel density estimator, which replaces the rigid box of the histogram with a smooth kernel function placed at each observation.

Given observations $x_1, x_2, \dots, x_n$, the kernel density estimator (KDE) at a point x is defined as

$$\hat{f}(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - x_i}{h}\right), \quad (2.3)$$

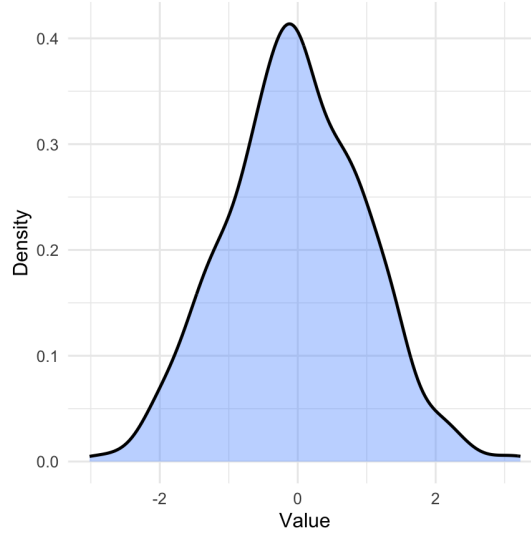


Figure 2.5: An example of a density plot.

where h is ‘the window width’ or ‘smoothing parameter’ which controls the degree of smoothing applied to the estimations. Intuitively, the estimate is constructed by placing a smooth ‘bump’ of shape K and width h at each observation and summing the contributions. The kernel K determines the shape of the bumps while h determines their width.

Though many options exist for the kernel function, a common choice is the normal kernel

$$K(u) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{u^2}{2}\right), \quad (2.4)$$

which ensures that $\hat{f}$ inherits the smoothness properties of K and is itself a valid probability density.

2.5.3 Violin Plot

The violin plot, introduced by Hintze and Nelson (1998) [4], combines the boxplot and density plot into one diagram. For each group, the kernel density estimate $\hat{f}(x)$ is computed as in equation 2.3 and then reflected symmetrically about a central vertical axis, producing a shape whose width at any value x is proportional to $\hat{f}(x)$.

When two groups are displayed side by side, differences in variance manifest as differences in the overall width and shape of the violins: a group with larger variance produces a violin with a broader profile and more prominent tails, while a group with smaller variance produces a more compact, spindle-like shape. Because the viewer must compare areas and widths rather than positions or lengths, violin plots engage what Cleveland and McGill (1984)[2] classify as lower-ranked perceptual tasks, which are generally associated with reduced accuracy relative to position-based comparisons. This has implications for the statistical power of the visual hypothesis test, as discussed in Section 4.

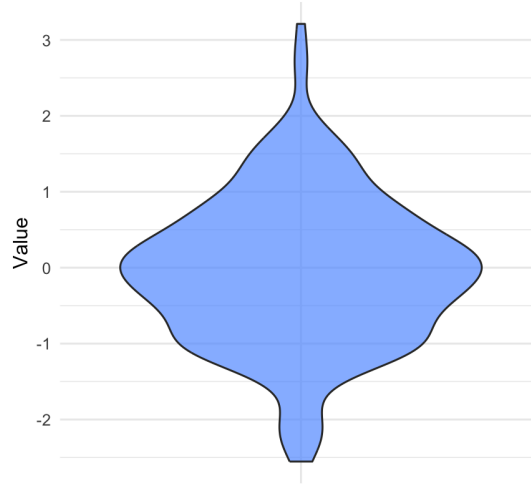


Figure 2.6: An example of a violin Plot.

2.5.4 Absolute Density Plot (Abs)

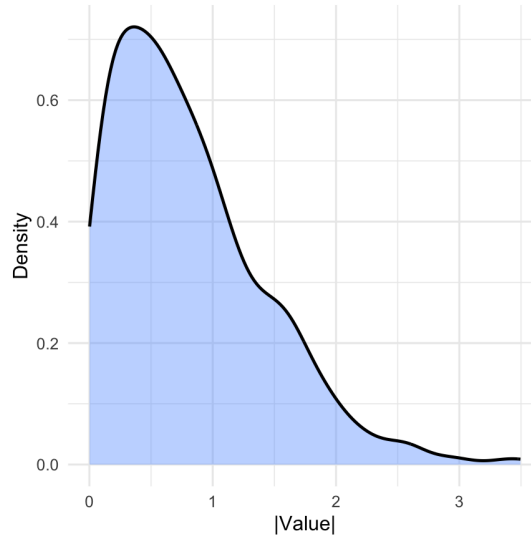


Figure 2.7: An example of an absolute density plot.

The absolute density plot is a transformation of the standard kernel density estimate designed to anchor variance comparisons to a common origin. Given observations $x_1, x_2, \dots, x_n$ from a distribution with mean μ , the data are first mean-centred and then transformed to their absolute values:

$$z_i = |x_i|, \quad i = 1, 2, \dots, n. \quad (2.5)$$

A kernel density estimate $\hat{f}_Z(z)$ is then computed on the transformed values z_i , in an analogous manner to the standard density plot.

Under this transformation, variance is directly encoded as the horizontal spread of the density away from the origin. A distribution with larger variance

produces a density that is more spread out along the positive real axis, while one with smaller variance produces a density concentrated near zero. When two groups are overlaid, the comparison reduces to one of relative horizontal position from a shared baseline, rather than a comparison of widths or areas. This anchoring to a common origin is consistent with the principle identified by Cleveland and McGill (1984)[2] that judgements of position along a common scale are the most accurately perceived of all elementary graphical tasks. Therefore, the absolute density plot represents an attempt to translate a scale-based inferential task into the most perceptually efficient encoding available.

2.6 Effect Size

Effect size refers to the magnitude of a difference or association observed in a study. As Sullivan and Feinn (2012) [8] explain, it may be expressed either as an absolute effect size, such as the raw difference between group means, or as a standardised measure that accounts for variability in the data. Standardised effect sizes are particularly useful when variables do not have an immediately interpretable scale, or when results need to be compared across studies using different measurements. For this reason, effect size is an essential complement to statistical significance when interpreting quantitative findings.

Chapter 3

Methods

3.1 Choice of Tests

We considered many hypothesis tests to assess for a visual experiment. The tests focused on meeting the following criteria:

- **Commonly used:** We wanted the goal of the experiment to be easily understood by other statisticians. This meant that the only new understanding required was related to the visual elements.
- **Clear graphical elements:** Since the goal of a visual test is to assess some visual cue, it is important that the hypotheses be clearly visualisable.
- **Clear null and alternative claims:** Precision in this area makes it easy to generate null data for the lineup protocol in order to ‘hide’ the true visualisation.

Initially, we considered three candidate hypotheses: the correlation between explanatory and response variables, the equality of means across two datasets, and the equality of variance between two datasets.

We constrained the synthetic data generation to a Gaussian distribution, $X \sim N(\mu, \sigma^2)$, with μ held constant at 0. This constraint was necessary to isolate variance as the sole independent variable, preventing mixing up other visual cues from acting as ‘distractors’ that could artificially alter detection accuracy and increase the risk of type III errors (correctly rejecting the null hypothesis, but for the wrong reasons). In practice, data rarely have equal means, so this assumption slightly hampers generalisability.

While all three meet the basic criteria, they offer different levels of graphical flexibility. Correlation is most effectively communicated via a scatterplot, but it is difficult to represent it using other common statistical plots. Similarly, testing for a difference in means (μ) often results in a simple comparison of location (e.g., the median line in a boxplot); however, this difference is often only perceptible when the effect size is large, regardless of the plot type used.

Ultimately, the test for equal variance was selected because of its graphical versatility. Unlike correlation, which is primarily suited to scatterplots, variance

can be encoded using several distinct geometric forms, specifically boxplots, density plots and violin plots. This variety is essential for our primary research goal: measuring how different visual encodings affect the power of a visual hypothesis test.

The selection of variance also allows for an investigation into the ‘Scale vs. Position’ perception gap. While most traditional plots require the viewer to compare the ‘Scale’ (width or area) of the data, our methodology tests whether transforming variance into ‘Position’ (as seen in the absolute density plot) yields a statistically significant increase in visual power, consistent with Cleveland’s hierarchy of graphical perception.

3.2 Choice of Visualisation

In designing a visual hypothesis test, careful consideration must be afforded to selecting the data visualisation that most accurately represents the null hypothesis. The chosen plot must clearly convey what we would expect under the null hypothesis so that deviations are easily identified by the viewer. For example, one would not use a scatterplot in order to test for equality of means, as this information is far easier to glean from a visualisation such as a boxplot. Further thought should be afforded to how accurately this information is presented to the viewer. Cleveland et al. [2] provide a rigorous framework for these decisions, outlining principles that enhance the accuracy and reliability of statistical inference derived from data visualisations.

3.2.1 Testing for Correlation

In our first lineup test, we aimed to identify a dataset exhibiting correlation from a lineup of null datasets with no correlation. We found that the scatterplot was the most effective visualisation for detecting the true dataset. By allowing viewers to compare points along a common scale, scatterplots make patterns of association immediately obvious, allowing the viewer to accurately identify correlated data as can be seen in Figure 3.1. In contrast, visualisations such as density plots or boxplots do not encode pairwise relationships directly, making it difficult to detect correlation with the eye. This advantage aligns with Cleveland et al.[2], who claim that visualisations supporting direct comparisons along common axes tend to yield higher accuracy in statistical inference.

3.2.2 Testing for Variance

Another classic hypothesis test we aim to recreate visually is that of equal variance (σ^2). When considering the variance of a dataset, our focus changes from association to spread. In order to create a rigorous visual hypothesis test, we require a plot that makes detecting variability in the data obvious and accessible. We considered several candidates: boxplots, density plots, violin plots and an absolute density plot. A boxplot summarises spread through the interquartile range and whiskers, allowing the viewer to make length comparisons along

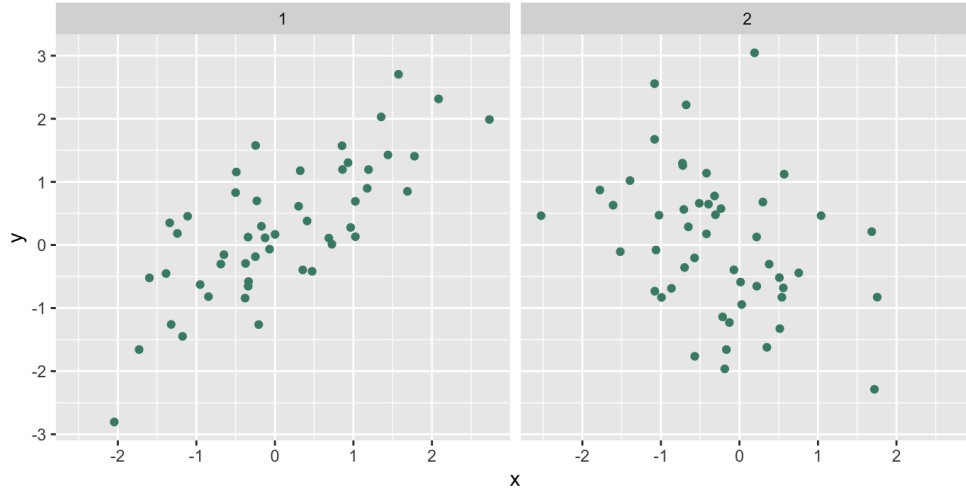


Figure 3.1: Correlated plot vs non-correlated plot.

a common vertical scale. However, boxplots hide the shape of a distribution beyond quartiles and may conceal subtle differences in tail behaviour and modal bulges. A standard density plot conveys the full distribution, including skewness and modes, and the ability to directly overlay two densities makes comparison more accurate for the viewer. Violin plots rely on comparisons of width, which correspond to area perception rather than direct positional comparison, which can reduce precision when the task is specifically to compare variance.

The absolute density plot was hypothesised to be particularly effective because variance is translated into horizontal spread from a common origin, enhancing perceptivity by anchoring all groups to the same reference point. One should note that each plot has different strengths for different test-statistics. For example, the boxplot would be more suitable for testing equality of means, since the location of the mean is readily available in the visualisation.

3.3 Construction of Lineups

In order to construct lineups, we had to generate several synthetic datasets under the null hypothesis of equal variance. This was done in a series of steps.

- **True Data Generation:** Firstly, we generated a synthetic dataset containing data in two groups. Group A was generated with a standard deviation of 1 and group B was generated with a standard deviation greater than 1. This could then be plotted to observe a difference in variance. To ensure the visual cue was detectable but not trivial, we utilised two specific effect sizes of $\sigma = 1.2$ and $\sigma = 1.7$. These values represent a weaker and a stronger departure from the equal-variance null, allowing us to compare plot performance in different variance. This allowed us to assess the threshold at which different visual encodings begin to fail as the variance ratio approaches 1:1.

- **Null Data Generation:** Secondly, we created the null datasets by permuting the group labels. This created 19 datasets with no defined relationship between the variables. This was done making use of the Nullabor[1] package in R, which contains functions which can permute data in this way. The use of permutation as opposed to simply simulating new data from a normal distribution is an important choice. Permutation ensures that the null plots contain the exact same marginal distributions and sampling noise as the true data plot. This prevents the true data from standing out due to outliers or incidental artefacts, ensuring that a correct selection by the participant is a result of detecting the variance difference specifically and avoiding Type III errors.
- **Visualisation:** Finally, we visualised the lineup using one of our four visualisation techniques, with the true dataset randomly hidden within the nineteen null datasets.

To maintain the integrity of the visual test, the location of the true plot was randomized across a 4×5 grid for every lineup. This randomization prevents ‘positional bias’, where participants might instinctively look toward the center or corners of the grid.

We used this pipeline to generate a lineup for each visualisation at standard deviation levels of 1.2 and 1.7 in order to retrieve data on how the visualisation impacts inference at different levels of difference.

3.4 Experiment

3.4.1 Survey Design

We constructed an online survey to assess participants’ ability to identify the true plot within a lineup. The survey was administered via Google Forms. Before completing the survey, the respondents gave some information about their experience with statistical software and their level of statistical training. These aspects allow us to investigate the impact of expertise on performance without prior knowledge or assumption that participants have experience in identifying the true plot. Participants were able to change their answers as they went due to the software used which could introduce biases if participants changed their mind after seeing statistical artefacts in later lineups.

Each question was designed to assess both the participants ability to find the plot and also the confidence of the participant in their ability. We included a null test to assess the participants engagement wherein we re-shuffled the datasets while preserving the relationship between the variables, and also changed the colour of the lines representing the groups. If the participants were approximately as successful on both lineups, then we would have evidence they were truly engaging with the survey.

In order to encourage the participants to complete the survey we decided to have ‘easier’ lineups at the beginning, where the standard deviation was higher in the true dataset. This was intended to mitigate frustration and make the

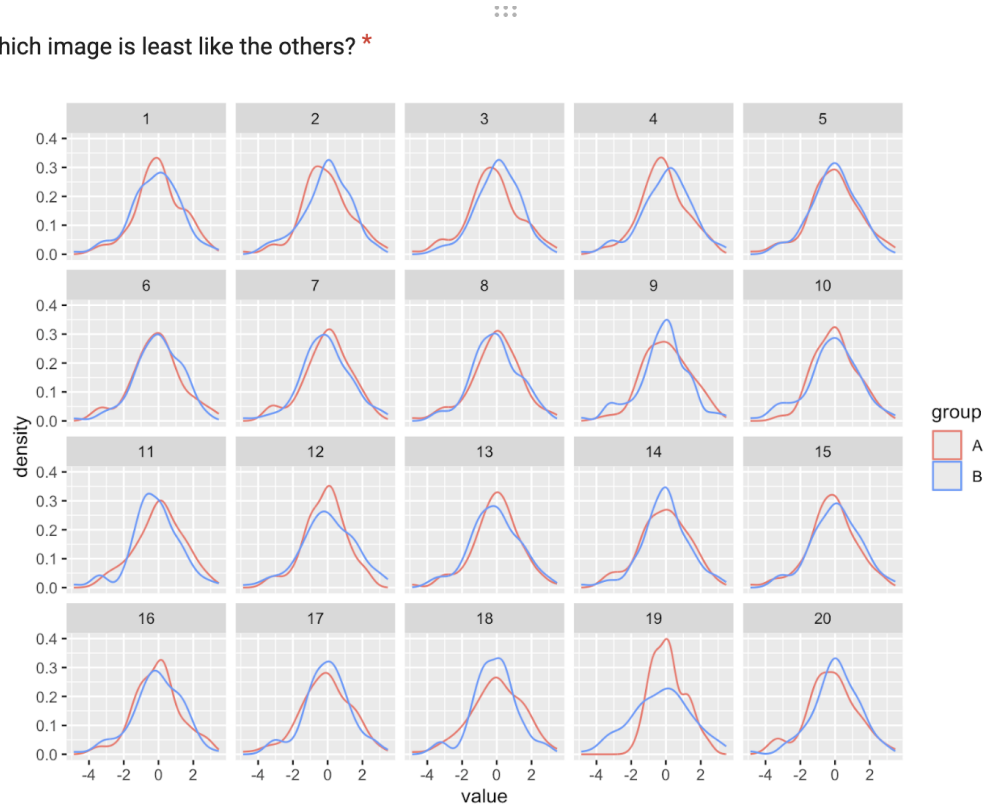


Figure 3.2: An example survey question.

participant more likely to answer every question. The survey then concluded with more challenging lineups, with the lower standard deviation difference.

3.4.2 Survey Structure

Participants were shown a sequence of lineups generated from three commonly used visualisation types: density plots, boxplots and violin plots. Each lineup contained one true plot and 19 null plots displayed in 4×5 grid.

For each lineup, participants were asked to select the plot they believed was ‘least like the others,’ corresponding to the plot they perceived as containing a real difference between Group A and Group B.

The final question of the survey invited the participants to provide a brief explanation describing the visual indicator that informed their choices throughout the survey. This would allow us to assess the cues in the various visualisations that made them more or less suitable for the application.

3.4.3 Recruitment and Administration

Participants were recruited from the School of Mathematics, including academic staff (e.g., lecturers and professors) as well as postgraduate and undergraduate students. Recruitment was conducted through the internal mailing list, informal academic networks and personal contacts. All the participation was voluntary and anonymous and individuals could complete the survey at their convenience.

Each participant completed the survey only once and they did not receive any feedback on the accuracy of their choices, to avoid incorporating bias for future questions.

3.5 Logistic Regression

3.5.1 Logistic Regression Model

In the analysis of our survey results, we fit a logistic regression model, following the framework of Hosmer and Lemeshow (2013) [5], to predict the probability of correctly detecting the true dataset. Since detection is a binary outcome, this framework allows the probability of a correct response to be modelled as a function of explanatory variables such as plot type, year of study, confidence level and the standard deviation used to generate the data.

The logistic regression model expresses the conditional probability of a correct response, $\pi(\mathbf{x}) = \Pr(Y = 1 \mid \mathbf{x})$, through the logit transformation:

$$g(\mathbf{x}) = \ln\left(\frac{\pi(\mathbf{x})}{1 - \pi(\mathbf{x})}\right) = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p, \quad (3.1)$$

where the logit $g(\mathbf{x})$ is linear in the parameters and may range over the entire real line, while $\pi(\mathbf{x})$ is constrained to the interval $(0, 1)$. The inverse of the logit transformation recovers the fitted probability:

$$\hat{\pi}(\mathbf{x}) = \frac{e^{g(\mathbf{x})}}{1 + e^{g(\mathbf{x})}}. \quad (3.2)$$

Model parameters are estimated by maximum likelihood, which yields values of β that maximise the log-likelihood

$$L(\beta) = \sum_{i=1}^n \left\{ y_i \ln[\pi(\mathbf{x}_i)] + (1 - y_i) \ln[1 - \pi(\mathbf{x}_i)] \right\}. \quad (3.3)$$

Differentiating $L(\beta)$ with respect to β_0 and β_1 and setting the resulting expressions equal to zero yields the likelihood equations

$$\sum_{i=1}^n [y_i - \pi(x_i)] = 0 \quad (3.4)$$

and

$$\sum_{i=1}^n x_i [y_i - \pi(x_i)] = 0. \quad (3.5)$$

3.5.2 Assumptions of the Logistic Regression Models

Although logistic regression is well suited to this task of binary classification, several assumptions and practical limitations should be considered when interpreting

the fitted models.

- **Binary Response variable:** Logistic regression requires the response variable to be binary. In this study, the outcome variable records whether a participant correctly identified the true panel in a lineup (correct or incorrect). This satisfies the binary response requirement of the model.
- **Independent Observations:** The logistic regression models that were conducted assume that each observation is independent. In our survey, each participant answered multiple questions to identify which plot is the most different from the others. This means that the observations might be correlated with the participants. Thus, the independence assumption is only approximately satisfied.
- **No Multicollinearity:** Logistic regression assumes that explanatory variables are not highly correlated with each other. This assumption is primarily relevant in models that contain multiple predictors, such as those that include both the plot type and the standard deviation. Therefore, multicollinearity should be assessed for the fitted models, in order to conclude whether it is satisfied.
- **Linearity in the Logit:** The model assumes a linear relationship between the predictors and the logarithmic odds of the outcome. In our analysis, most predictors are categorical variables. Thus, this assumption is satisfied by using indicator variables that represent each category.
- **Large Sample Size:** Logistic regression requires a sufficiently large sample size to produce valid conclusions from the fitted values. The number of responses to our survey was relatively small with 47 participants, but each participant evaluated multiple lineups, making the total number of observations larger.

Chapter 4

Survey Results and Exploratory Analysis

4.1 Survey Overview

As noted in Section 3.4, the survey began with some questions about the participants' statistical experience and stage of study. The participants were then presented with an example lineup experiment in order to help them understand the goals of the survey. Finally, the participants moved on to the 9 lineup experiments before being prompted to describe what visual cues assisted them in making their decision.

The survey received 48 total submissions with 47 answering all lineup questions.

While the survey was designed to provide a structured evaluation of visual hypothesis testing, several practical limitations must be considered when interpreting the results. Due to time and financial constraints, the study utilised a relatively small sample size of $n = 47$ participants, with each individual answering 9 questions. Consequently, the total number of observations is limited, which reduces the overall statistical power of the analysis.

Because of this restriction, the findings presented in this report may not be fully robust and might not reflect the patterns that would emerge in a larger, more representative sample. Specifically, the small sample size increases the margin of error for our estimates and makes it difficult to detect subtle interactions between variables. Therefore, while the results offer valuable exploratory insights into how plot types and standard deviation influence identifiability, they should be treated as a 'proof of concept' rather than a definitive conclusion. Further research with a larger participant pool would be required to validate these trends and ensure their generalisability across different groups.

4.2 Survey Result and Statistical Evidence

The participant composition indicates a primarily undergraduate audience with a small amount of postgraduate representation. The most advanced statistics were separated between Pre-Honours and Honours (both 41.7%), with 14.6%

completing an MSc and 2.1% completing a PhD. The most prevalent years of study among respondents were Year 4 (35.4%) and Year 2 (27.1%), with 16.7% in the MSc category and a small number in other categories. The experience level of R varied from ‘never used R’ (29.2%) to ‘use R regularly’ (20.8%), with the largest group having completed a formal course (39.6%). The skew towards undergraduate respondents is likely to hinder our ability to conclude much about the effect of statistical training on the individual’s power as a test-statistic.

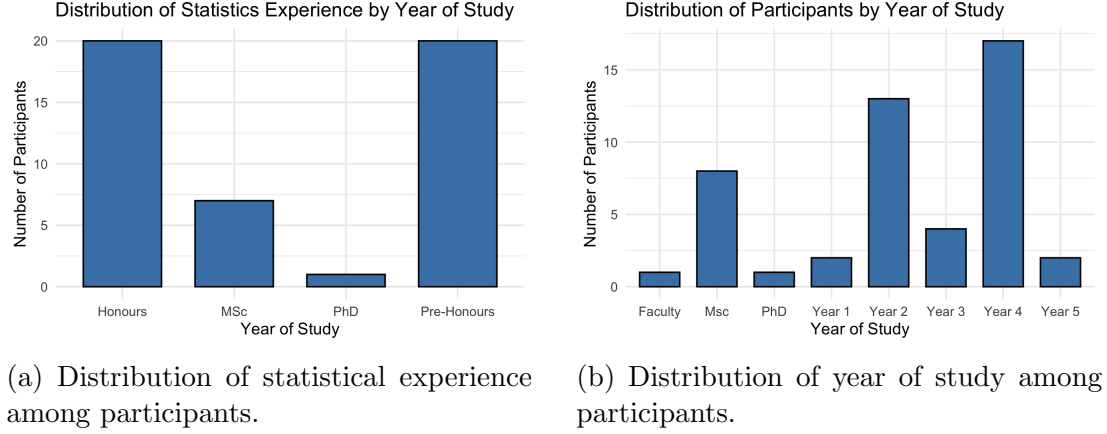


Figure 4.1: Participant Distributions.

A fundamental outcome is the detection rate, which is the percentage of participants who selected the correct panel. Table 4.1 summarises the lineup-level results, including accuracy, mean confidence, and the chance-based tail probability. The ($p_0 = 1/20$) baseline is consistent with the standard lineup interpretation, which equates to ($p \approx 0.05$) for a single correct selection in a 20-panel lineup.

We included an attention check to ensure participants were genuinely engaging with the survey. This involved replicating a density plot lineup with the panel order shuffled and the group colours inverted. The true plot was detected 97.9% of the time in the original lineup and 93.6% of the time in the altered lineup. A Fisher’s Exact Test confirmed that this difference was not statistically significant ($p = 0.62$), suggesting that participants were neither relying on positions nor superficial colour cues. Instead, responses appear to reflect genuine visual inference based on distributional shape, which strengthens confidence in the validity of the broader survey results.

4.2.1 Chance-Based

Each lineup contained $m = 20$ panels, so probability of selecting the true panel with no visual aid is $p_0 = 1/m = 0.05$. For a given lineup with n responses and k correct identifications, the chance-based probability is computed under the null model

$$X \sim \text{Binomial}(n, p_0)$$

with the right-tail probability

$$P(X \geq k) = \sum_{x=k}^n \binom{n}{x} p_0^x (1 - p_0)^{n-x} = 1 - P(X \leq k - 1).$$

This value quantifies the probability of observing at least k correct responses solely by chance. Smaller values indicate stronger evidence that participants performed above random guessing for that lineup.

Table 4.1: Lineup-level summary of responses, accuracy, confidence, and chance-based tail probabilities under random guessing ($p_0 = 1/20$).

Lineup	Plot family	True panel	Resp.	Correct	Acc.	Mean conf.	Chance-based $P(X \geq k)$
1	Violin	2	47	40	85.1%	3.17	4.0×10^{-45}
2	Density	19	46*	44	95.7%	4.20	5.3×10^{-55}
3	Boxplot	5	47	19	40.4%	2.45	3.4×10^{-13}
4	Density (abs)	5	47	46	97.9%	4.19	6.4×10^{-59}
5	Density (abs)	10	47	44	93.6%	4.13	7.9×10^{-54}
6	Violin	2	47	6	12.8%	2.28	2.9×10^{-2}
7	Density	6	47	31	66.0%	3.09	3.2×10^{-29}
8	Boxplot	1	47	3	6.4%	2.00	4.20×10^{-1}
9	Density (abs)	7	47	25	53.2%	3.19	1.5×10^{-20}

* One response was missing/invalid for this lineup and was excluded from the analysis.

4.2.2 Investigating Predictors

We aim to investigate possible relationships between our explanatory variable, whether or not the participant was able to find the correct plot, and our various explanatory variables. We first conducted some exploratory data analysis to find the mean accuracy for various factors.

		Plot Type	Mean Accuracy
Standard Deviation	Mean Accuracy	Abs	0.816
1.2	0.346	Box	0.234
1.7	0.825	Density	0.806
		Violin	0.489

Table 4.2: Mean classification accuracy by confidence level and plot type.

Table 4.2 demonstrates that the participants in the survey were more accurate on the greater effect size, indicating that this could be a useful predictor.

In addition, Table 4.2 shows our hypothesis that the type of plot used to conduct visual hypothesis testing has a great impact on the ability of the participant to detect the true dataset. The absolute density plot and density plot both demonstrate high mean detection accuracy, with violin plots being detected around half of the time and boxplots performing the most poorly with only 23% mean accuracy. The detection of the boxplots is still far higher than random

selection, which indicates that there is still a small amount of useful information being observed in the boxplots, relating to our null hypothesis of equality of variance.

4.2.3 Exploratory Patterns From the Survey

We visualise the relationship between correctness and a number of critical variables prior to constructing regression models. Each plot displays the mean accuracy within each group, as indicated by a large coloured point.

A significant dependence on plot family is indicated by Figure 4.2.

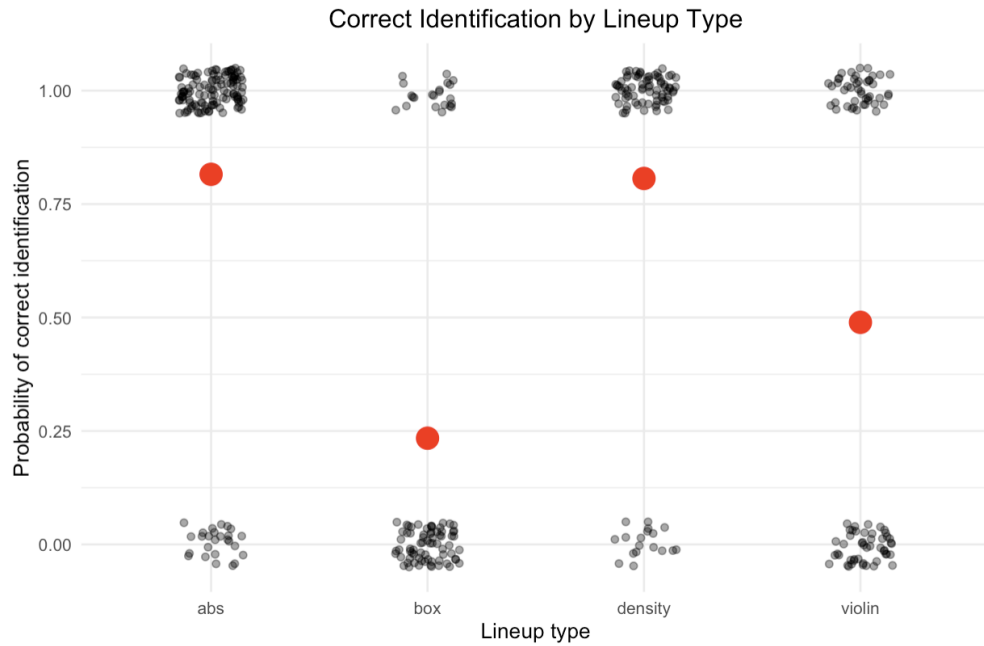


Figure 4.2: Accuracy by lineup type.

The absolute density plots (abs) and density plots both demonstrate high average accuracy. Conversely, violin plots exhibit intermediate performance, and boxplots perform significantly worse. This provides preliminary evidence that the choice of visual encoding affects the power of the lineup task.

A clear monotone relationship between self-reported confidence and accuracy is illustrated in Figure 4.3

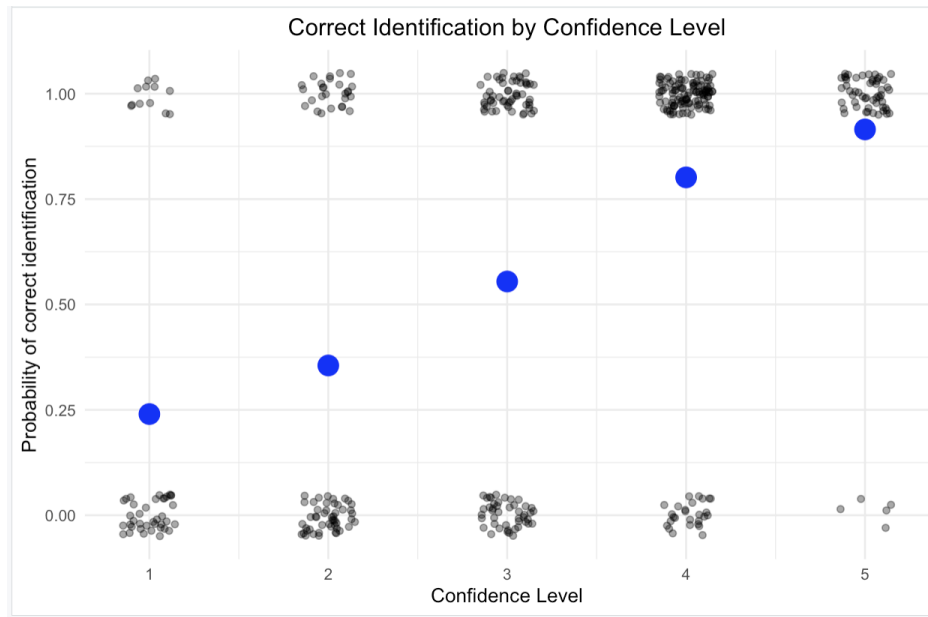


Figure 4.3: Accuracy by confidence level.

In particular, higher confidence levels are associated with higher probabilities of correct identification. This suggests that the confidence of participants contain valuable information regarding the perceived intensity of the signal and the difficulty of the task.

Then Figure 4.4 illustrates relatively small variations among year-of-study categories. The sample sizes for several categories are small, so these differences should be interpreted cautiously, despite the fact that some groups (e.g., MSc/PhD) do appear slightly higher. This encourages the use of environmental variables as secondary predictors and priority of plot type and confidence.

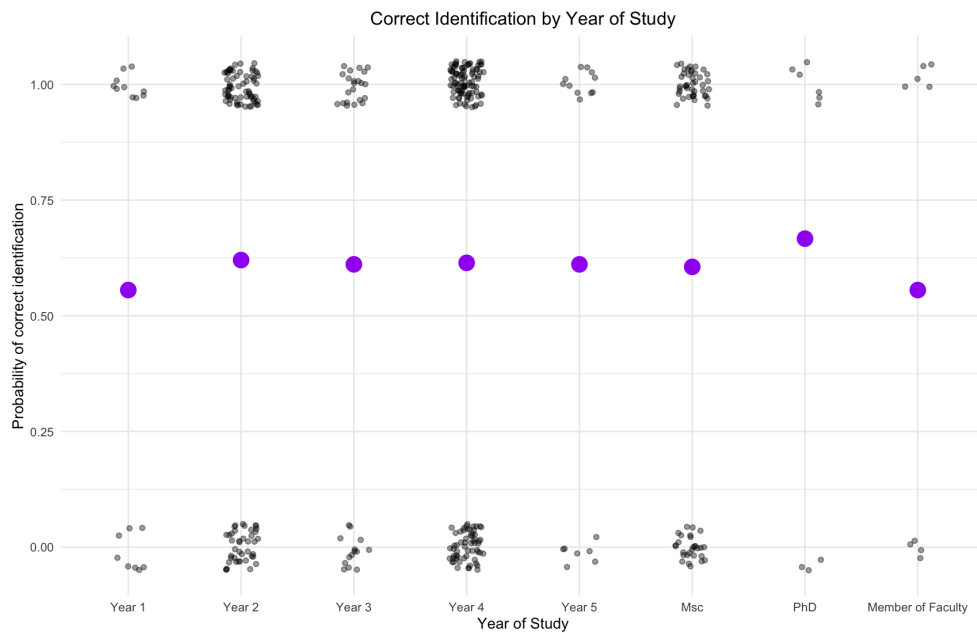


Figure 4.4: Accuracy by year of study.

In addition to participant-level factors, our lineup design also adjusted the variability of the null plots that were employed to generate each lineup. True datasets were constructed under two standard deviation settings ($\sigma \in \{1.2, 1.7\}$). For each plot type (abs, box, density, violin), Figure 4.5 summarises the mean accuracy (with 95% confidence intervals) as a function of null-plot variability. The dashed horizontal line indicates chance performance ($p_0 = 1/20 = 0.05$). The accuracy of all four plot types was higher when the null variability was larger ($\sigma = 1.7$) than when it was smaller ($\sigma = 1.2$). Specifically, for abs it increased from 53% to 96%, for the box from 6% to 40% (with $\sigma = 1.2$ approximately equal to chance), for the density from 66% to 96%, and for the violin from 13% to 85%. In general, the true plot is more distinguishable as a result of an increase in null variability, which in turn enhances the detection performance of participants.

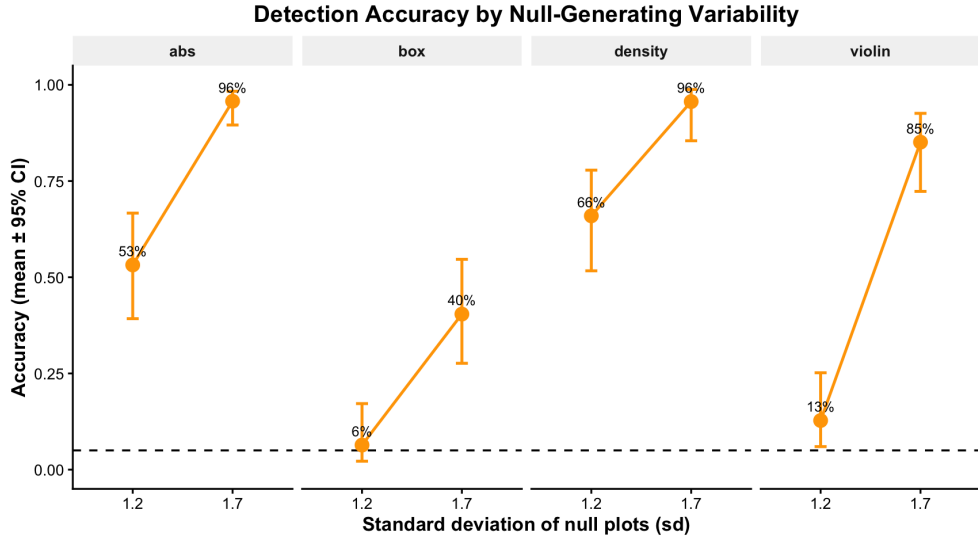


Figure 4.5: Detection accuracy by null-generating variability ($\sigma \in \{1.2, 1.7\}$) for each plot type. Points show means with 95% confidence intervals; the dashed line indicates chance accuracy ($p_0 = 1/20 = 0.05$).

The findings of this chapter indicate that the effectiveness of the lineup protocol is significantly influenced by the visual encoding of the underlying signal. Throughout the survey, density-based displays attained the highest detection accuracy, while boxplots exhibited the lowest performance overall, and violin plots held an intermediate position. We observed that performance enhanced when the difference in standard deviation became greater, and that participants' confidence generally aligned with accuracy.

The next chapter moves from description to explanation. It analyses the qualitative evidence found in participants' open-ended responses and relates these reported cues with the quantitative patterns detected in estimated power, confidence, and logistic regression. Our objective is to demonstrate not only certain plot types exhibited superior performance but also to identify the perceptual and statistical attributes that may induce these differences.

Chapter 5

Qualitative and Quantitative Analysis

This chapter does not focus on the contrasting results of visualisations, but on the reasons behind the observed differences. In the context of visual inference, this question is particularly significant, as the observer does not simply provide a correct or incorrect answer, but rather reacts to particular visual cues that render one plot more distinguishable than the others.

To solve this issue, we combine two types of evidence. Initially, we analyse participants' open-ended comments to identify the visual cues that influenced their decisions, including spread, peak structure, overlap or apparent separation between groups. Secondly, we connect these qualitative reports with the quantitative results presented later in the analysis, including estimated visual power, confidence patterns and logistic regression models. This enables us to analyse our findings on two levels simultaneously: the perceptual level, by inquiring which cues were most prominent to observers, and the statistical level, by examining if those cues correspond with higher modelled probabilities of accurate identification.

5.1 Open-Ended Responses: Visual Cues Reported by Participants

We proceed to analyse the survey's open-ended responses to the question: 'What details in the visualisations informed your decisions?' in order to investigate the subjective motivations for the results. This offers qualitative evidence at the participant level regarding the visual signals that were most prominent during the identification of the lineup.

We separated participants into groups based on detection accuracy. Higher accuracy participants had an accuracy rate above the median(0.67) and lower accuracy participants had an accuracy below and equal to the median. Table 5.1 shows the results of the first 10 participants.

Participant	Accuracy	Accuracy group
2	0.667	Lower accuracy
3	0.667	Lower accuracy
4	0.667	Lower accuracy
5	0.667	Lower accuracy
6	0.375	Lower accuracy
7	0.778	Higher accuracy
8	0.778	Higher accuracy
9	0.778	Higher accuracy
10	0.444	Lower accuracy
11	0.556	Lower accuracy
⋮	⋮	⋮

Table 5.1: Participant-level detection accuracy and grouping based on the median split.

We found that the most frequently used terms by participants were *separation/overlap*, *center/quantiles*, and *shape/pattern*. This implies that the lineup selection was influenced by the visually obvious differences among groups, the central positioning, and the overall distributional form.

We also observed that the type of cue used varied among the performance groups. Participants with high accuracy were relatively more likely to refer to *spread/variance* and, in some cases, more specific distributional features. In contrast, participants with lower accuracy referred to *separation/overlap* and *shape/pattern* more frequently. This suggests that the observation of a difference in the appearance of one panel may not always be sufficient for accurate identification; instead, higher-performing participants may have depended more on the structural properties or characteristics of the distribution, such as variability or spread, rather than on the more immediate overall visual contrast alone.

Participants who mentioned signals such as *spread/variance*, *shape/pattern*, or *peak/density* demonstrated moderate to relatively high average accuracy. In contrast, themes such as *separation/overlap* were associated with a comparatively lower average accuracy in spite of their relatively high confidence. This is particularly intriguing because it implies that certain visual strategies may produce confidence without necessarily enhancing accuracy. Conversely, signals that exhibit more sensitivity to distribution appear to correlate more consistently with successful detection. Despite the fact that some categories were mentioned by a limited number of participants, these differences should still be regarded with caution. However, the overall pattern is consistent with the idea that cue selection is a significant factor in lineup performance.

5.2 Boxplot vs Violin Plot Comparison

In order to improve the aggregate results in Figure 4.5, Figure 5.1 illustrates two representative boxplot lineups under varying true-generating variability paramete-

ters. With restricted variability ($\sigma = 1.2$), the medians, interquartile ranges and outlier patterns of most panels are extremely similar, which results in boxplots that perform similarly to chance. The panel that looks the least like the others is easier to identify and the accuracy is improved when the between-panel variation becomes clearer (e.g. in whiskers, spread and outlier patterns) under increased variability ($\sigma = 1.7$). This qualitative comparison supports the trend shown in Figure 4.5, demonstrating that the accuracy of boxplots improves with an increase in true variability.

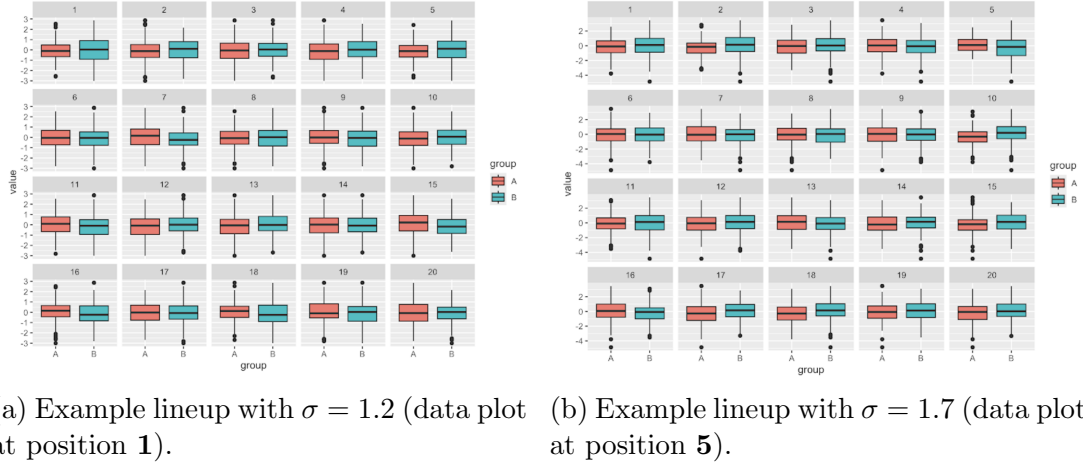


Figure 5.1: Representative boxplot lineups used to illustrate the effect of null-generating variability. For clarity, the position of the data plot is stated in the caption (this position was not revealed to participants during the task).

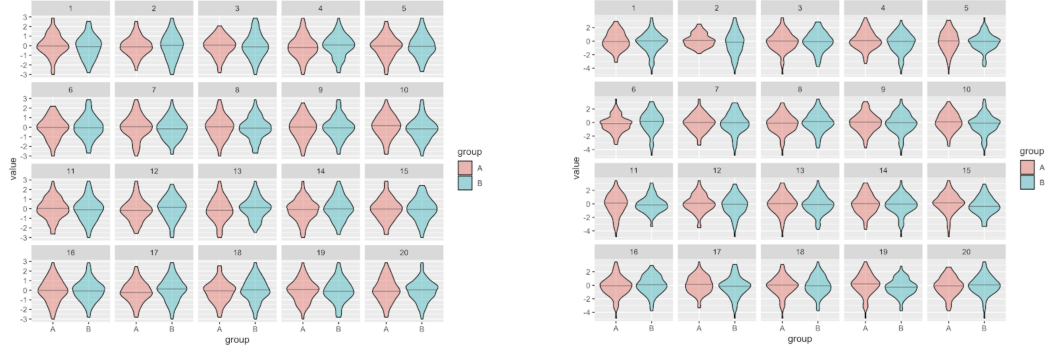
Figure 4.5 illustrates that the detection accuracy of violin plots improves significantly as the null-generating variability increases from $\sigma = 1.2$ to $\sigma = 1.7$ (13% to 85%). Figure 5.2 illustrates the two representative violin lineups under the two variable situations to provide a visual explanation for this trend. For clarity, the position of the data was not revealed to participants during the survey.

When $\sigma = 1.2$, most of the panels exhibit similar violin shapes for both groups, with only slight differences in width or position. Violin plots depend on the comparison of distributional shapes (as seen in Chapter 2). Hence, their compressed and closely resembling outlines offer minimal visual separation, resulting in the data plot not distinctly contrasting with the null plots. This aligns with the decreased accuracy noted at $\sigma = 1.2$.

Conversely, when $\sigma = 1.7$, the distributions display increased dispersion, form characteristics more visually apparent including tail thickness, asymmetry. The amplified and diverse signals help the detection of the most unusual panel, explaining the significant improvement in the accuracy of the violin plot.

Disparity in detection performance provides compelling evidence for the impact of visualisation on statistical power. Despite being similar tools for distributional assessment, they provide vastly different results when tasked with detecting variance disparities in a lineup protocol.

At the lower standard deviation level ($\sigma = 1.2$) both plots perform poorly, with the boxplot giving a near chance detection rate of 6% and the violin displaying



(a) Example violin lineup with $\sigma = 1.2$ (data plot at position **2**). (b) Example violin lineup with $\sigma = 1.7$ (data plot at position **2**).

Figure 5.2: Representative violin lineups under different null-generating variability settings. The data-plot position is shown for the reader but was not revealed to participants during the task.

a slightly better 13%. This suggests that at low effect sizes, both distributions lack sufficient visual cues to assist the viewer to distinguish the true data from the null plots.

At the higher standard deviation level ($\sigma = 1.7$), we observe a significant increase in the gap between the two visualisations. Violin plots experience a substantial increase to 85% accuracy-aligning closely with the high performance of density plots (96%)-while boxplots only reach a modest 40%. This disparity could arise due to a number of different factors.

We can see the violin plots presented to the participants in Figure 5.2b. Panel two displays an extremely anomalous plot, in that it almost entirely lacks a tail below zero. This makes it an obvious candidate for a plot that looks ‘least like the others’ and thus it was selected frequently by the participants. Participants cited ‘cutoff points’ and ‘things like whether the shapes had long tails or were stumpy’. These criteria make the second panel the easy choice for participants. The same data is presented in a boxplot in Figure 5.1b, with the true data being displayed in panel five. It is immediately more difficult to discern differences in the plots. The two groups in panel five clearly have different spread, with the red plot displaying far narrower interquartile ranges. Participants frequently cited outliers as impacting their choices in the boxplot lineups. 29.8% of participants incorrectly selected panel fifteen in Figure 5.1b. Group A has a higher number of outliers than all other plots, a visual cue that is not apparent in violin plots, which may have impacted the decision of many participants.

The plots also utilise different visual elementary perceptual tasks as researched by Cleveland et al. [2] In a boxplot, the viewer is primarily performing a task of judgment of relative position and length. To detect a difference in variance, the participant must compare the vertical distance between the first and third quartiles (the box) or the total reach of the whiskers. These are linear comparisons. However, the presence of outliers (as seen in panel fifteen of Figure 5.1b) introduces a ‘distractor’. Because outliers are represented by distinct points, they draw

the eye more effectively than the subtle compression of a box’s length, leading to the high false-discovery rate (29.8%) observed in our results.

In contrast, the violin plot engages the viewer in judgments of area and curvature. The ‘stumpy’ tails and ‘cutoff points’ cited by participants are changes in the geometric shape of the plot rather than just its linear extent. The symmetry of the violin plot provides a visual signal, the change in width occurs on both sides of the central axis, which emphasises the visual evidence of a variance shift compared to a single boxplot whisker.

Furthermore, the dramatic increase in accuracy for violin plots at $\sigma = 1.7$ suggests there is a threshold where the signal of the standard deviation overcomes the difficulties associated with comparing area and curvature. Once the standard deviation disparity is large enough to noticeably alter the ‘bulge’ or ‘tail’ of the violin plot, it moves from a task of subtle estimation to one of pattern recognition. The boxplot, by reducing the distribution to five summary points, fails to provide this shape change, forcing the user to continue assessing area and linear distances, which is easily disrupted by the visual noise of outliers.

It is also possible that the large difference in the accuracy for the two plots at the larger standard deviation difference is simply due to random chance. As visible in the plot, the data generated did not contain points far below zero, resulting in a short tail. This feature is likely to be what drew participants to the plot, however this is not a universal trait of violin plots, and other datasets generated in the same manner will not have this visual cue. Hence, future research would need to be conducted to investigate whether or not this difference is universal, or data specific.

Consequently, for the specific task of identifying variance disparities, the violin plot appears to serve as a higher-power ‘test statistic’ than the boxplot, particularly as the effect size increases.

5.3 Estimated Visual Test Power

The previous subsection provides a qualitative explanation for the differences between the performance of boxplots and violin plots. However, it is important to quantify how effective each visualisation is as a statistical test within the lineup framework. To formalise on that idea, we estimated the power of the lineup test under different effect sizes.

In order to quantify the uncertainty in the estimated probability of correct identification, a 95% Wilson confidence interval [10] was computed for the binomial proportion. The lower and upper bounds of this interval were then substituted into the same binomial power calculation to obtain the corresponding lower and upper bounds for the estimated power.

Figure 5.3 shows the estimated power for the lineup test for each different type of visualisation used in our survey, across the two different effect sizes ($\sigma = 1.2$ and $\sigma = 1.7$). The dashed horizontal line shows the rejection probability if one were to guess randomly, which is equal to 0.05 for a lineup consisting of 20 plots.

Therefore, we use Figure 5.3 to obtain several clear patterns. Firstly, we observe that both the density plots and the absolute density plot (abs) achieve very

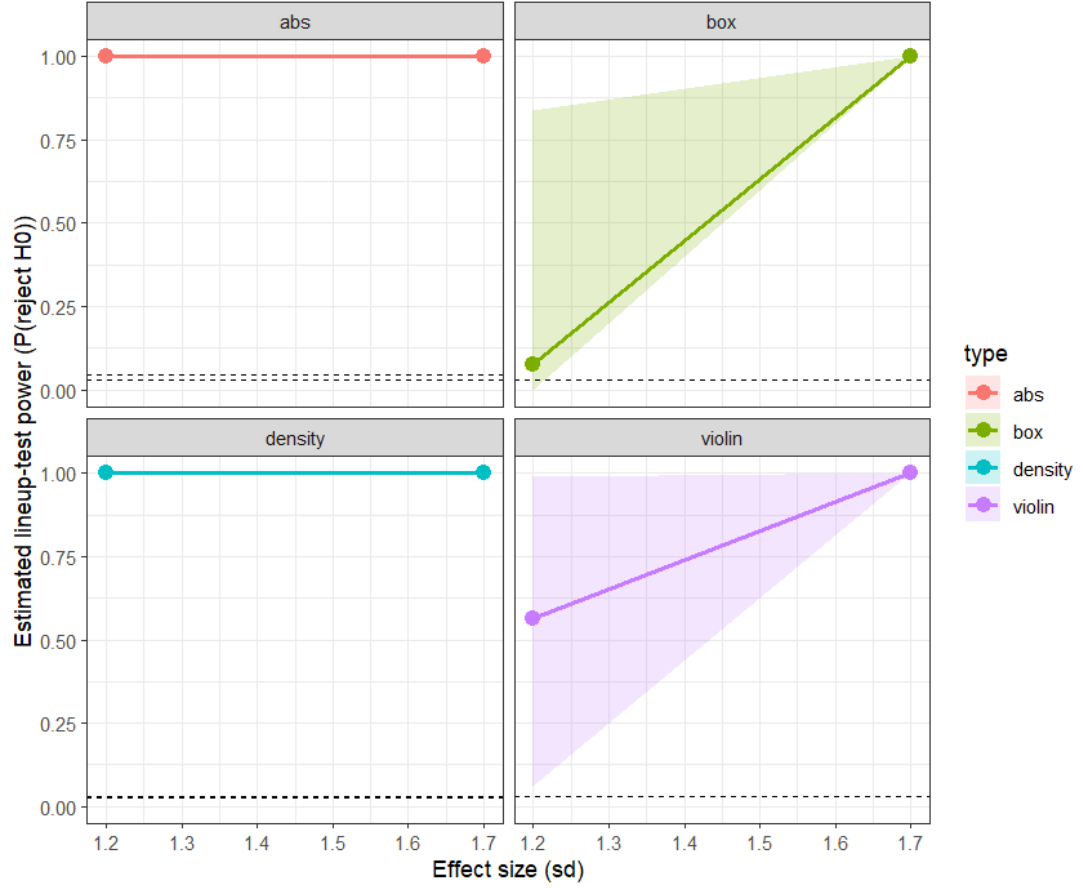


Figure 5.3: Estimated lineup test power for the two effect sizes and each visualisation type.

high estimated power across both effect sizes. This suggests that the participants were able to identify the true data plot consistently when those density plots were used to represent the data.

In contrast, boxplots and violin plots do not perform as effectively as the density plots. We observe that violin plots show moderate power (0.56) when the effect size is at its smallest ($\sigma = 1.2$), but their performance improves considerably while the effect size increases. When σ approaches 1.7 the power is nearly perfect.

Observing the estimated power of the boxplots we can say that they perform relatively poorly when the effect size is at its lowest, with power values close to the baseline level (dashed line). Although their performance improves significantly while the effect size increases (approaching the perfect power), they still remain less effective than the other types of visualisations.

This estimation of power is consistent with the pattern observed before in the detection accuracy analysis. Although these results summarise the performance of each type of visualisation, they do not account for other factors that may influence a participant's response. Hence, to examine these factors more formally, we model the probability of a correct response using logistic regression.

5.4 Baseline Logistic Regression: Effect of Plot Type

Since the response variable is binary, we model the probability of correct identification using logistic regression. The omission of feedback during the survey aimed to prevent participants from learning biases as they proceeded throughout the survey, in order to try and preserve some independence between answers. Furthermore, each participant will have ingrained perceptual biases which inform their choices. Hence, the following logistic regression analysis is used to investigate broad trends. Let p denote the probability that a given response is correct. The baseline model includes plot type as a categorical predictor:

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1\mathbb{I}(\text{type} = \text{box}) + \beta_2\mathbb{I}(\text{type} = \text{density}) + \beta_3\mathbb{I}(\text{type} = \text{violin}),$$

with `abs` as the reference category.

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	1.48684	0.21716	6.847	7.55×10^{-12}
as.factor(type)box	-2.67246	0.32634	-8.189	2.63×10^{-16}
as.factor(type)density	-0.05972	0.34066	-0.175	0.861
as.factor(type)violin	-1.52940	0.29955	-5.106	3.30×10^{-7}

Table 5.2: Logistic regression coefficients for plot type.

Table 5.2 reports the fitted coefficients. Relative to `abs`, boxplots and violin plots have significantly lower log-odds of correct identification (both p-values ≤ 0.001), while density plots are statistically indistinguishable from `abs` ($p = 0.861$). This is consistent with Figure 4.2, where `abs` and density display similar mean accuracies. We also calculated the Akaike Information Criterion (AIC) for the model. The AIC is a metric used to decide between models based on their fit to the data. For this model, the AIC was calculated to be 466.74. A lower AIC indicates a better model fit.

To interpret the effect sizes, we convert log-odds to predicted probabilities via the inverse-logit function:

$$\hat{p} = \frac{\exp(\eta)}{1 + \exp(\eta)}, \quad \eta = \beta_0 + \beta_k.$$

This yields $\hat{p}_{\text{abs}} \approx 0.815$, $\hat{p}_{\text{density}} \approx 0.806$, $\hat{p}_{\text{violin}} \approx 0.489$, and $\hat{p}_{\text{box}} \approx 0.234$, matching the empirical means in Table 4.2. In odds-ratio terms, boxplots reduce the odds of correct identification by a factor of $\exp(-2.672) \approx 0.069$ relative to `abs`, while violin plots reduce the odds by $\exp(-1.529) \approx 0.217$.

5.5 Effect of Academic Experience: Analysis by Year of Study

To investigate whether a participant’s level of academic progression influences their ability to correctly identify the data plot, a logistic regression model was fitted using both plot type and year of study as categorical predictors. The model formula is defined as:

$$\begin{aligned} \log\left(\frac{p}{1-p}\right) = & \beta_0 + \beta_1\mathbb{I}(\text{type} = \text{box}) + \beta_2\mathbb{I}(\text{type} = \text{density}) + \beta_3\mathbb{I}(\text{type} = \text{violin}) \\ & + \beta_4\mathbb{I}(\text{year} = \text{Year 1}) + \beta_5\mathbb{I}(\text{year} = \text{Year 2}) \\ & + \beta_6\mathbb{I}(\text{year} = \text{Year 3}) + \beta_7\mathbb{I}(\text{year} = \text{Year 4}) \\ & + \beta_8\mathbb{I}(\text{year} = \text{Year 5}) + \beta_9\mathbb{I}(\text{year} = \text{MSc}) + \beta_{10}\mathbb{I}(\text{year} = \text{PhD}). \end{aligned}$$

This resulted in the output in table 5.3;

Variable	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	1.184	0.788	1.502	0.133
Type: Boxplot	-2.678	0.327	-8.193	2.55×10^{-16}
Type: Density Plot	-0.060	0.341	-0.176	0.860
Type: Violin Plot	-1.532	0.300	-5.109	3.23×10^{-7}
Year: MSc	0.286	0.818	0.350	0.726
Year: PhD	0.619	1.117	0.554	0.579
Year: Year 1	0.000	0.941	0.000	1.000
Year: Year 2	0.353	0.802	0.440	0.660
Year: Year 3	0.301	0.863	0.349	0.727
Year: Year 4	0.320	0.792	0.403	0.687
Year: Year 5	0.301	0.948	0.318	0.751

Table 5.3: Logistic regression results for the effect of plot type and year of study on correct identification.

We can see in table 5.3 that the most significant predictors remain the plot types. Specifically, boxplots ($p < 2.55 \times 10^{-16}$) and violin plots ($p = 3.23 \times 10^{-7}$) show a strong, statistically significant negative effect on the probability of a correct identification compared to the baseline (absolute density plots). Interestingly, none of the ‘year of study’ levels reached statistical significance at the 5% level. The p -values for MSc (0.726), PhD (0.579), and various undergraduate years (ranging from 0.660 to 1.000) suggest that higher levels of academic experience do not necessarily translate to higher accuracy in this specific visual task.

This suggests that the visual cues provided by the choice of visualisation are far more influential than the formal statistical training of the observer. Even participants in advanced stages of study (PhD/MSc) were subject to the same visual limitations imposed by less effective encodings like boxplots. This aligns with the report’s earlier assertion that the choice of visual representation is a primary factor in the power of a visual hypothesis test.

The data for this section is limited by the scale of the study. Due to the small number of postgraduate and faculty responses, the standard errors for the MSc and PhD categories are relatively high (0.818 and 1.117 respectively), leading to non-significant p-values.

In addition, we assessed the multicollinearity among the explanatory variables using variance inflation factor (VIF [3]) diagnostics. Since the predictors in the model are categorical variables with multiple degrees of freedom, the function reports generalised variance inflation factors (GVIF). This was done to check whether the assumption of no multicollinearity is satisfied.

Predictor	GVIF	Df	$GVIF^{1/(2Df)}$
type	1.001961	3	1.000327
year	1.001961	7	1.000140

Table 5.4: Generalised variance inflation factors (GVIF) for the logistic regression model including plot type and year of study..

Table 5.4 presents the generalised variance inflation factors for the predictors included in the model. The values for both the type of the plot and year of study are approximately equal to 1. Values close to 1 indicate that the explanatory variables are not strongly correlated, thus, the assumption is satisfied.

5.6 Extended Logistic Regression: Incorporating Standard Deviation

Building on the baseline model, we investigated the effect of the standard deviation (σ) used to generate the true dataset. In addition to plot type, the standard deviation was included as a categorical factor to determine how the magnitude of the signal affects identification accuracy. This results in the following extended additive model:

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 \mathbb{I}(\text{type} = \text{box}) + \beta_2 \mathbb{I}(\text{type} = \text{density}) + \beta_3 \mathbb{I}(\text{type} = \text{violin}) + \beta_4 \mathbb{I}(\sigma = 1.7)$$

In this model, ‘Density (abs)’ and $\sigma = 1.2$ serve as the reference categories for plot type and standard deviation, respectively. The model results are summarised in Table 5.5.

The results show that increasing the standard deviation from 1.2 to 1.7 has a highly significant positive effect on the log-odds of correct identification ($\beta_4 = 2.991, p < 0.001$). This provides strong evidence that all plot types effectively communicate variance differences, making them suitable for visual hypothesis testing.

The AIC for this model was calculated as 348.751. This is significantly lower than the previous model and provides further evidence that the standard deviation level has a strong impact on the identifiability of the true dataset.

	Estimate	Std. Error	z value	Pr(< z)
(Intercept)	0.1245	0.2669	0.466	0.641
as.factor(type)box	-3.3989	0.4301	-7.902	2.74×10 ⁻¹⁵
as.factor(type)density	0.4653	0.3877	1.200	0.230
as.factor(type)violin	-1.6954	0.3801	-4.460	8.18×10 ⁻⁶
as.factor(σ)1.7	2.9991	0.3353	8.944	<2×10 ⁻¹⁶

Table 5.5: Logistic regression coefficients for plot type and standard deviation (σ).

This is also a multiple logistic regression model, and therefore we need to check if the multicollinearity assumption is being satisfied. Thus, we use the VIF diagnostics [3] to assess multicollinearity.

Predictor	GVIF	Df	$GVIF^{1/(2 \times Df)}$
type	1.5127	3	1.0714
σ	1.5127	1	1.2299

Table 5.6: Variance Inflation Factor (VIF) results for the logistic regression model.

Table 5.6 shows the values recorded after the Variance Inflation Factor diagnostics. The values for both the type of the plot and the effect size (σ) are close to 1, which indicates that the explanatory variables are not correlated. Therefore, the assumption of multicollinearity is satisfied.

To investigate whether some visualisations are more sensitive to changes in standard deviation than others, we introduced an interaction term between plot-type and σ. This creates the following model:

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1\mathbb{I}(\text{type} = \text{box}) + \beta_2\mathbb{I}(\text{type} = \text{density}) + \beta_3\mathbb{I}(\text{type} = \text{violin}) \\ + \beta_4\mathbb{I}(\sigma = 1.7) + \beta_5\mathbb{I}(\text{type} = \text{box})\mathbb{I}(\sigma = 1.7) \\ + \beta_6\mathbb{I}(\text{type} = \text{density})\mathbb{I}(\sigma = 1.7) + \beta_7\mathbb{I}(\text{type} = \text{violin})\mathbb{I}(\sigma = 1.7).$$

We investigated the joint impact of standard deviation and plot-type using ANOVA. The output is as follows,

Model Term	Df	Deviance	Resid. Df	Resid. Dev	Pr(>Chi)
NULL			421	563.90	
as.factor(type)	3	105.16	418	458.74	< 2 × 10 ⁻¹⁶
as.factor(σ)	1	119.99	417	338.75	< 2 × 10 ⁻¹⁶
as.factor(type):as.factor(σ)	3	2.77	414	335.98	0.4285

Table 5.7: Analysis of deviance table for interaction model.

The non-significance of the interaction term indicates that the increase in identifiability from having a higher standard deviation is felt equally across all visualisations. This indicates an independence between the difficulty of the task

of identification, and the visualisation used to communicate the data. Despite some visualisations clearly having superior cues for detecting the variance, the gap between the best and worst visualisations remains consistent across the standard deviation levels.

The ordering of plot types is not an artefact of a particular effect size, but rather a stable property of the visualisations themselves. Therefore, a researcher selecting a visualisation for a variance-based hypothesis test does not need to consider the expected magnitude of the effect: the relative superiority of density-based encodings holds regardless of effect size.

This result is also theoretically consistent with Cleveland and McGill’s hierarchy of graphical perception tasks [2]. The perceptual task required by each visualisation does not change as the standard deviation increases. Only the magnitude of the visual signal changes. It is therefore expected that the hierarchy of encodings remains fixed across effect sizes, with the interaction term absorbing negligible variance as a result.

However, this conclusion should be treated with some caution. With only two effect sizes tested, the statistical power to detect a genuine interaction was limited. A non-significant interaction term in this setting does not definitively rule out the possibility that the ranking could change at more extreme values of σ . For instance, this can be obtained at very small effect sizes where all plots approach chance performance, or at very large effect sizes where variance difference are unmistakable. A wider range of effect sizes would be needed to establish the generalisability of this finding more rigorously.

5.7 Analysis of Participant Confidence

A final factor considered was the subjective confidence of participants. Although logistic regression models analyse objective accuracy, the confidence metric provides insight into the perceived clarity of the visual evidence and the persuasive power of different visualisations. As shown in our results, at the $\sigma = 1.2$ level, average confidence was notably low for both boxplots (2/5) and violin plots (2.3/5). In contrast, density and absolute density plots elicited much higher initial confidence, at 3.1/5 and 3.2/5 respectively. As the standard deviation improved at $\sigma = 1.7$, confidence in violin plots increased significantly to 3.2/5, while boxplot confidence saw a much more marginal increase to 2.4/5. The participants were also very confident in assessing the density and absolute density plots at this higher signal level, with those values both increasing to 4.2/5. This leads to several conclusions:

- **Calibration:** Participants are generally well-calibrated. They correctly identify when a task is more difficult (low σ) and adjust their confidence accurately across all plot types.
- **The Supremacy of Density Encodings:** The high confidence scores for density and absolute density plots suggest that showing the full distribution is the most effective way to communicate statistical differences. The fact

that these plots reached a confidence of 4.2/5 at $\sigma = 1.7$ indicates that they provide the most visual evidence to an observer.

- **Violin Plots as a Middle Ground:** Violin plots occupy a middle ground. While they share the density-based approach of the density plots, their lower confidence scores compared to standard density plots suggest that the mirrored, vertical orientation may be slightly less intuitive for participants to decode quickly, though they still remain far superior to boxplots.
- **Impact on Visual Power:** These results suggest that density-based plots possess higher explanatory power. In a real-world scenario, a researcher is more likely to act on a visual finding if the visualisation provides the level of detail necessary to feel confident in the observation.

5.8 Successes of Density-Based Methods

The most consistent finding across both the logistic regression models and the participant confidence is the overwhelming superiority of density-based visualisations over traditional summary-based methods like the boxplot. Unlike boxplots, which reduce a dataset to a five-point summary, density plots reveal the full distribution of the data, allowing observers to identify nuances such as changes in spread or the presence of multimodal distributions. Absolute density plots proved particularly effective by achieving the highest confidence ratings of 4.2/5 at the $\sigma = 1.7$ level. By providing a common baseline for comparison, these plots minimise the cognitive load required to judge the distance between group means, effectively increasing the visual power of the test. This finding aligns with those of Cleveland et al. [2], with the absolute density converting the assessment to that of location on a common scale, the highest-ranked task in the hierarchy, which humans perceive with the greatest accuracy. In contrast, traditional density plots or violin plots require the viewer to assess lengths or areas, which are lower-ranked tasks prone to higher error rates.

A critical success of density methods is the alignment between objective performance and subjective power. Although participants' evaluations of the boxplots were both inaccurate and uncertain, their evaluations of the density plots were significantly more likely to identify the correct plot and feel justified in their choice. As noted in Section 4, the jump in confidence for density-based methods as signal strength improved was nearly double that of the boxplot. In conclusion, for visual hypothesis testing regarding differences in standard deviation, and possibly other distributional attributes, density-based methods are the most robust as they provide the highest probability of a correct rejection of the null hypothesis while ensuring the researcher can trust the visual evidence presented, a quality essential for reliable exploratory data analysis.

Chapter 6

Conclusion

This report has investigated the application of the lineup protocol as a formal framework for visual hypothesis testing, bridging the gap between exploratory data analysis and rigorous statistical inference. By positioning the human observer as a test statistic, we have demonstrated that visual interpretation can be systematised to provide inferential validity comparable to traditional numerical methods. All code and data used in the project are available at the GitHub repository in appendix [A](#).

6.1 Summary of Findings

The primary finding of this study is that the choice of visual encoding fundamentally determines the statistical power of a visual hypothesis test. Our experimental results revealed a stark contrast in detection accuracy across different plot types:

- **Density and Absolute Density Plots:** These visualisations achieved the highest performance, with mean accuracies of approximately 82% and 81% respectively. The ‘absolute’ density plot, which centres observations at zero, was particularly effective by translating variance into horizontal spread from a common origin, thereby facilitating more direct visual comparison.
- **Violin Plots:** This type of plot showed intermediate performance with a mean accuracy of roughly 49%. While providing more distributional information than boxplots, the reliance on area perception rather than direct positional comparison may have reduced its precision for variance testing.
- **Boxplots:** These performed most poorly, with a mean accuracy of 23%. While still exceeding the 5% random chance baseline, boxplots often concealed the subtle differences in tail behaviour and modal bulges necessary for identifying variance shifts.

Furthermore, we found that the strength of the statistic we are testing for has a strong impact on accuracy. The boxplot in particular improved from a near-baseline 6% detection rate to 40%, confirming that visual cues gain power with the effect size in the same manner as numerical tests.

6.2 Implications and Limitations

The results presented in this report indicate that practitioners must carefully consider and select visualisations depending on the specific hypothesis being tested. For variance-based hypotheses, methods using density plots should be preferred, since they consistently outperformed other visualisations. Boxplots are traditional for comparing means, but are less suitable for variance-based visual inference compared to methods based on density plots.

The findings also reinforce Cleveland and McGill's[2] findings, with the effectiveness of the visualisations aligning with their hierarchy of elementary perception tasks. This provides support for their ideas in the context hypothesis testing.

However, these findings should be viewed more as a proof of concept rather than hard evidence. The study was conducted on a small sample of 47 people, primarily students from the University of Edinburgh School of Mathematics with statistical education. This is not a representative sample, which limits the generalisability of the findings to the general public. The skew towards undergraduate students reduces the validity of the null result in student experience, a result that requires further investigation.

The short duration of the survey was designed to reduce survey fatigue in the participants and ensure they completed the survey. However, this reduces the amount of information we can collect about a participant and their choices.

6.3 Future Research

The findings of this project provide useful information on the effectiveness of different visualisation techniques for visual hypothesis testing. Despite that, these findings open several avenues for further investigation.

6.3.1 Survey Design and Implementation

The small sample size of the survey is the main detractor from our findings. Testing a larger and more diverse sample of participants would lend more credibility to the findings. Therefore, future research could involve a larger number of participants across a broader range of backgrounds.

Furthermore, another potential improvement would be the change in the number of lineups each participant evaluated. In our survey, each participant evaluated nine lineups. Increasing the number of lineups to which each participant was exposed would help assess the strongest factors affecting identification, since it would provide a larger number of observations. In particular, a set of Rorschach tests could be used to calculate each participant's Type I error rate (the probability of rejecting the null hypothesis when it is true). This was omitted from our survey to avoid survey fatigue and encourage engagement.

In order to combat positional bias, where certain panels may be preferred over others due to their position in the lineup, the survey could be conducted using eye-tracking technology, to assess where participants are drawn naturally.

More investigation should be done into the aspects of the visualisations which observers focus on most through open-ended questions and mining the answers for frequently cited words.

6.3.2 Exploring Additional Hypothesis Tests

The time constraints of the project allowed us to focus only on the hypothesis test for equal variance across two samples. However, the lineup protocol can be applied to a much wider range of statistical hypotheses.

Ideally, future research would consider other, potentially less commonly used hypothesis tests and investigate whether our findings carry over. This would involve selecting a hypothesis, finding or creating a visualisation or visualisations which effectively convey the data feature important to the hypothesis, and finally surveying a large sample of participants. This study focused on boxplots, violin plots, density plots, and absolute density plots when evaluating differences in variance. However, many other graphical representations could be investigated within the lineup framework.

6.3.3 Expanding the Range of Effect Sizes

Another important extension would involve improving the experimental design of the lineup tasks. In this study, only two levels of standard deviation were considered ($\sigma \in \{1.2, 1.7\}$). This allowed us to investigate how the accuracy of the correct identification varies between a relatively weak and a stronger signal.

Future research can investigate a wider range of different effect sizes. This would make the interpretation of the results easier and more understandable. Furthermore, it would allow researchers to construct more detailed power curves for different visualisations.

In addition to varying effect sizes, future research should also consider lineups in which the true dataset is generated under the same conditions as the null plots (in our case the same σ). In such cases, there is no true signal and any identification of a plot showing the true dataset would correspond to a Type I error (the probability of rejecting the null hypothesis when it is true).

6.3.4 Mixed-Effects Logistic Regression

Finally, future research can explore different statistical modelling approaches. The logistic regression models that we used in this study were applied under the assumption that every observation is independent. However, in our survey, each participant evaluated nine different lineups. Hence, the responses of each individual might be correlated due to either a different statistical level or a different general ability to identify unusual patterns in the plots. This does not fully satisfy the assumption of the independence in observations.

Thus, an important extension of this analysis would be the use of mixed-effects logistic regression models. In this framework, participant-level random effects are included to account for variability between individuals while still estimating the

overall influence of explanatory variables such as the type of the plot and the standard deviation used to generate the data.

This approach can help determine more reliably how much variation in lineup performance is due to differences between the type of visualisations and effect sizes and how much variation is due to the differences between participants. This would help to understand better the factors that influence the resulting performance of visual hypothesis testing.

Appendix A

R Code and Survey Data

All R code and survey data is stored in the GitHub repository at <https://github.com/MDuncan13/VHTProject.git>

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